6.8 Graphs of Polar Equations

Name _____ Period ____ Date _____

Pre-Calculus 6.8 (30-minute plan) · Student sheet

I can…

- test a polar equation for symmetry.
- find the zeros and maximum |r| of a polar equation.
- recognize and sketch limaçons and rose curves.

Vocabulary · fill in the missing word

Limaçon — the _____ of r = a ± b cos θ or r = a ± b sin θ

Cardioid — a _____çon with a = b (heart-shaped)

Rose curve — the _____ of r = a cos nθ or r = a sin nθ, n ≥ 2

Lemniscate — the _____ of $r^2 = a^2 \sin 2\theta$ or $r^2 = a^2 \cos 2\theta$

Part 1 · Notes and examples (fill in each note, then work the example)

Symmetry tests: replace (r, θ) with (r, −θ) for the polar axis, (r, π − θ) for the line θ = π/2, and (r, θ + π) or (−r, θ) for the _____.

Example 1

Test r = 3 + 2 cos θ for symmetry.

Zeros and maximum |r|: find the θ-values where r = 0 and where |r| is _____.

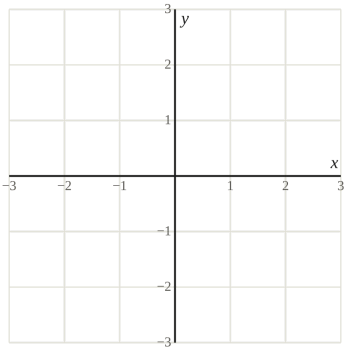
Rose curves r = a cos nθ or a sin nθ: n petals if n is odd, _____ petals if n is even; each petal is |a| long.

Example 2

Describe each rose curve.

a. r = 2 cos 3θ

b. r = 3 sin 2θ



Limaçons r = a ± b cos θ: a/b < 1 inner loop; a/b = 1 _____; 1 < a/b < 2 dimpled; a/b ≥ 2 convex.

Example 3

Classify each limaçon.

a. $r = 2 + 2 \sin \theta$

b. $r = 1 + 2 \cos \theta$

c. $r = 3 + 2 \cos \theta$

d. $r = 5 + 2 \cos \theta$

Part 2 · Partner practice

Partner practice

Classify $r = 3 - 3 \sin \theta$, give its symmetry, and find its maximum $|r|$.